

Height Wrangling Practice

Robert Thomas Kohn

10/1/25

Question: Does Sex Impact Ones Height?

```
library(Lock5Data)
data(HeightData)
head(HeightData)
```

```
##   ID Sex Year_1 Year_1.25 Year_1.5 Year_1.75 Year_2 Year_3 Year_4 Year_5 Year_6
## 1  1  M   81.3    84.2    86.4    88.9    91.4   101.1   109.5   115.8   121.9
## 2  2  M   76.2    80.4    83.2    85.4    87.6    97.0   104.6   112.3   118.9
## 3  3  M   76.8    79.8    82.6    84.7    86.7    94.2   100.4   107.1   112.3
## 4  4  M   74.1    78.4    82.6    85.4    88.1    98.6   104.4   111.0   116.3
## 5  5  M   74.2    76.3    78.3    80.3    82.2    89.4    96.9   104.1   110.7
## 6  6  M   76.8    79.1    81.1    84.4    87.4    94.0   102.4   109.2   116.1
##   Year_7 Year_8 Year_8.5 Year_9 Year_9.5 Year_10 Year_10.5 Year_11 Year_11.5
## 1  130.0  138.2   141.1  144.3   147.5   150.5   153.4   156.2   159.7
## 2  125.0  130.1   133.0  135.4   137.5   139.7   142.2   144.2   146.2
## 3  118.6  124.0   126.5  128.9   131.2   133.4   135.8   138.4   141.0
## 4  123.2  129.9   133.0  136.0   138.7   141.4   144.0   146.4   148.8
## 5  115.8  121.7   125.2  128.5   131.2   134.2   137.4   140.5   143.3
## 6  121.9  127.8   129.8  132.4   135.0   137.3   139.7   142.0   144.1
##   Year_12 Year_12.5 Year_13 Year_13.5 Year_14 Year_14.5 Year_15 Year_15.5
## 1   163.8   168.8   174.9   181.2   186.3   189.6   191.3   192.1
## 2   148.1   149.8   151.6   153.8   156.3   159.2   163.3   167.7
## 3   143.6   146.8   150.8   155.1   159.5   163.3   166.8   167.8
## 4   151.4   154.5   157.7   162.2   167.4   172.5   176.3   178.5
## 5   145.9   148.7   152.0   155.8   160.4   165.1   168.5   170.4
## 6   146.9   150.0   153.4   157.3   161.6   165.2   168.8   170.1
##   Year_16 Year_16.5 Year_17 Year_17.5 Year_18
## 1   192.8   193.2   193.8   194.3   195.1
## 2   171.5   174.3   176.1   177.4   178.7
## 3   168.8   169.8   170.9   171.2   171.5
## 4   179.8   180.7   181.4   181.6   181.8
## 5   171.6   172.4   172.5   172.5   172.5
## 6   171.5   172.1   172.6   173.2   173.8
```

```
summary(HeightData)
```

```
##      ID      Sex      Year_1      Year_1.25      Year_1.5
## Min.   : 1    F:54   Min.   :67.30   Min.   :72.30   Min.   :74.90
## 1st Qu.:24    M:39   1st Qu.:72.80   1st Qu.:76.80   1st Qu.:80.40
## Median :47                Median :74.90   Median :78.70   Median :82.30
## Mean   :47                Mean   :74.77   Mean   :78.69   Mean   :82.11
```

```
## 3rd Qu.:70          3rd Qu.:76.80  3rd Qu.:80.70  3rd Qu.:84.20
## Max. :93          Max. :82.20  Max. :88.90  Max. :90.90
## Year_1.75        Year_2        Year_3        Year_4
## Min. :78.40      Min. :80.90  Min. : 88.10  Min. : 94.7
## 1st Qu.:83.40    1st Qu.:86.00  1st Qu.: 94.50  1st Qu.:101.9
## Median :85.40    Median :88.10  Median : 96.00  Median :103.8
## Mean :85.15      Mean :87.91  Mean : 96.29  Mean :103.8
## 3rd Qu.:87.00    3rd Qu.:89.70  3rd Qu.: 98.20  3rd Qu.:105.8
## Max. :93.70      Max. :97.30  Max. :106.30  Max. :114.4
## Year_5          Year_6          Year_7          Year_8
## Min. :100.1      Min. :105.7  Min. :112.5  Min. :117.0
## 1st Qu.:108.7    1st Qu.:114.9  1st Qu.:121.6  1st Qu.:126.8
## Median :110.7    Median :117.6  Median :124.1  Median :129.9
## Mean :110.8      Mean :117.6  Mean :123.9  Mean :129.8
## 3rd Qu.:112.8    3rd Qu.:120.1  3rd Qu.:126.5  3rd Qu.:133.3
## Max. :123.5      Max. :130.6  Max. :137.7  Max. :144.4
## Year_8.5        Year_9          Year_9.5        Year_10
## Min. :119.2      Min. :121.4  Min. :124.5  Min. :126.8
## 1st Qu.:129.8    1st Qu.:132.8  1st Qu.:135.3  1st Qu.:137.8
## Median :132.9    Median :135.6  Median :138.5  Median :141.4
## Mean :132.7      Mean :135.7  Mean :138.5  Mean :141.4
## 3rd Qu.:136.4    3rd Qu.:139.5  3rd Qu.:142.5  3rd Qu.:145.0
## Max. :148.4      Max. :152.5  Max. :156.6  Max. :161.5
## Year_10.5       Year_11          Year_11.5       Year_12
## Min. :129.0      Min. :131.6  Min. :133.5  Min. :135.6
## 1st Qu.:140.5    1st Qu.:143.7  1st Qu.:146.3  1st Qu.:148.6
## Median :144.2    Median :147.0  Median :150.5  Median :153.8
## Mean :144.4      Mean :147.5  Mean :150.7  Mean :153.9
## 3rd Qu.:148.5    3rd Qu.:152.0  3rd Qu.:155.3  3rd Qu.:159.0
## Max. :166.1      Max. :170.2  Max. :173.5  Max. :176.0
## Year_12.5       Year_13          Year_13.5       Year_14
## Min. :137.6      Min. :139.5  Min. :141.8  Min. :145.0
## 1st Qu.:151.6    1st Qu.:154.4  1st Qu.:158.1  1st Qu.:160.5
## Median :157.0    Median :159.2  Median :161.6  Median :163.6
## Mean :157.0      Mean :159.9  Mean :162.6  Mean :165.0
## 3rd Qu.:162.1    3rd Qu.:165.3  3rd Qu.:167.6  3rd Qu.:169.8
## Max. :177.3      Max. :178.3  Max. :181.8  Max. :186.3
## Year_14.5       Year_15          Year_15.5       Year_16
## Min. :148.2      Min. :150.8  Min. :152.1  Min. :152.3
## 1st Qu.:161.8    1st Qu.:163.0  1st Qu.:164.3  1st Qu.:164.4
## Median :166.1    Median :167.9  Median :168.6  Median :169.5
## Mean :167.0      Mean :168.6  Mean :169.7  Mean :170.6
## 3rd Qu.:172.1    3rd Qu.:172.6  3rd Qu.:174.7  3rd Qu.:176.2
## Max. :189.6      Max. :191.3  Max. :192.2  Max. :193.0
## Year_16.5       Year_17          Year_17.5       Year_18
## Min. :152.6      Min. :153.2  Min. :153.5  Min. :153.6
## 1st Qu.:164.7    1st Qu.:165.1  1st Qu.:165.1  1st Qu.:165.2
## Median :170.2    Median :170.6  Median :170.8  Median :170.9
## Mean :171.2      Mean :171.6  Mean :171.8  Mean :172.1
## 3rd Qu.:178.0    3rd Qu.:178.5  3rd Qu.:178.5  3rd Qu.:178.7
## Max. :193.6      Max. :194.0  Max. :194.3  Max. :195.1
```

```
hdc <- HeightData %>% mutate(across(everything(), as.factor)) #Cleaning and Factor
hd<-hdc%>%pivot_longer(cols = -c(ID, Sex), names_to = "Year", names_prefix="Year_", values_to = "Height")
```

```
head(hd)
```

```
## # A tibble: 6 x 4
##   ID    Sex    Year Height
##   <fct> <fct> <dbl>  <dbl>
## 1 1      M      1      61
## 2 1      M    1.25    100
## 3 1      M    1.5     128
## 4 1      M    1.75    102
## 5 1      M      2     164
## 6 1      M      3     226
```

```
gf_point(Height ~ Year, data = hd, color=~Sex)%>% gf_facet_wrap(~ Sex)%>%gf_labs(x = "Age of Subject", y = "Height")
```

